team-leftside-undergrads

Midterm Evaluation: The Leftside Undergrads

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Repository: https://github.com/aishanomani/The_Leftside_Undergrads_Midterm **Level:** Undergraduate

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Part 1: Operational Databases, Data Warehousing, and ETL (20 points)

Section 1: Environment Setup (3 pts)

Task	Points	Score	Notes
1.1 Docker compose ps (5 services)	1	1	Documented
1.2 \dt for all three databases	1	1	Tables listed
1.3 pgAdmin screenshot (3 servers)	1	1	Screenshot present

Subtotal: 3/3

Section 2: Explore Operational Databases (4 pts)

Task	Points	Score	Notes
2.1 Sales schema + normalization response	1.5	1.5	Schema output and written response present
2.2 HR schema + relationships response	1.5	1.5	Schema output and relationships explanation present
2.3 Four queries with output	1	1	All four query outputs included

Subtotal: 4/4

Section 3: Data Warehouse Concepts (4 pts)

Task	Points	Score	Notes
3.1 Star schema \d for 5 tables	0.5	0.5	All five warehouse tables shown
3.2 Operational vs warehouse comparison	2	2	Addresses denormalization, structural differences, dim_date

Task	Points	Score	Notes
3.3 Facts vs dimensions identification	1.5	1.5	Correct identification with grain and example query

Subtotal: 4/4

Section 4: ETL Pipeline (5 pts)

Task	Points	Score	Notes
4.1 ETL script analysis	2	2	Covers all four sub-questions
4.2 Count verification	1	1	Counts shown and matching
4.3 New order through ETL	2	2	Full flow documented

Subtotal: 5/5

Section 5: Analytical Queries (4 pts)

Task	Points	Score	Notes
5.1 Three analytical queries (SQL + output)	3	3	All three queries with correct SQL and output
5.2 EXPLAIN ANALYZE comparison	1	1	Both outputs and written comparison present

Subtotal: 4/4

Part 1 Total: 20/20

Part 2: Sharding the Sales Database (20 points)

Section 1: Setup & Exploration (4 pts)

Task	Points	Score	Notes
1.1 Docker compose ps (6 services)	1	1	All services shown
1.2 Count rows in each shard	1	1	Correct counts documented
1.3 Uneven distribution analysis	2	2	Hot shard problem identified; alternative shard key proposed

Subtotal: 4/4

Section 2: Update ETL Pipeline (8 pts)

Task	Points	Score	Notes
2.1 Diagnose ETL failure	2	2	Correctly identifies connection failure and needed changes
2.2 Modified ETL script + output	4	4	Clean implementation with `extract_sales_from_shards()` function; includes dict-based deduplication for customers/products; properly merges shard data

Task	Points	Score	Notes
2.3 Verify warehouse data + new order	2	2	Counts verified; new order flow documented

Subtotal: 8/8

Section 3: Cross-Shard Challenges (4 pts)

Task	Points	Score	Notes
3.1 Cross-shard query + limitation	2	2	Both outputs shown; limitation documented
3.2 Sharding trade-off analysis	2	2	Balanced discussion of benefits and drawbacks

Subtotal: 4/4

Section 4: EXPLAIN ANALYZE (4 pts)

Task	Points	Score	Notes
4.1 SE shard EXPLAIN ANALYZE	2	2	Output shown
4.2 Warehouse EXPLAIN ANALYZE + response	2	2	Warehouse isolation explained

Subtotal: 4/4

Part 2 Total: 20/20

Part 3: Adding Cassandra for Real-Time Order Ingestion (20 points)

Section 1: Setup & Cassandra Exploration (4 pts)

Task	Points	Score	Notes
1.1 Docker compose ps (8 services)	1	1	All eight services documented
1.2 Nodetool status + DESCRIBE	1	1	Cluster status and schema shown
1.3 Table design comparison	2	2	Denormalization and query-first design explained

Subtotal: 4/4

Section 2: Query-First Design Analysis (4 pts)

Task	Points	Score	Notes
2.1 Valid Cassandra queries	2	2	Both queries with output
2.2 Invalid query + explanation	2	2	Error documented; partition key restriction explained

Subtotal: 4/4

Section 3: Write Orders to Cassandra (4 pts)

Task	Points	Score	Notes
3.1 Dual inserts + verification	2	2	Both inserts and verifications present
3.2 Dual-write consistency	2	2	Logged batches mentioned as solution

Subtotal: 4/4

Section 4: Update ETL Pipeline (4 pts)

Task	Points	Score	Notes
4.1 ETL analysis	2	2	Identifies import, new function, and what stays the same
4.2 Modified ETL + verification	2	2	Working Cassandra extraction with deduplication; `order_date.date()` conversion correct; warehouse verification shown

Subtotal: 4/4

Section 5: Polyglot Persistence Reflection (4 pts)

Task	Points	Score	Notes
5.1 Comparison table + response	2	2	Accurate table; CAP positioning correct
5.2 Cassandra as warehouse argument	2	2	Strong argument against with specific reasons

Subtotal: 4/4

Part 3 Total: 20/20

Part 4: Distributed SQL Database Presentation (40 points)

> **Note on scoring:** Full marks awarded for the in-class presentation delivery portion (36 points: Technical Accuracy, TechRetail Evaluation, Presentation Quality, Visual Aids, Q&A Handling, Equal Participation). While there was room for improvement on each presentation, the feedback given and discussions held during class provide overall guidance for areas of growth. Full marks reflect the lengthy delay between the presentations and these evaluations being returned.

Category	Points	Score	Notes
Technical Accuracy	8	8	Full marks.
TechRetail Evaluation	8	8	Full marks.
Presentation Quality	8	8	Full marks.
Visual Aids	6	6	Full marks.

Category	Points	Score	Notes
Q&A Handling	2	2	Full marks.
Equal Participation	4	4	Full marks.
Slide Content (/10)	—	**7**	**Citus (11 slides).** Solid technical accuracy: correctly identifies Citus as PG extension, coordinator/worker model, CP positioning, 2PC (not Raft/Paxos). Good comparison table mapping current problems to Citus solutions. **Weaknesses:** Architecture diagram on slide 4 appears to be a generic database hierarchy image, not an original Citus-specific diagram (rubric requires original). Claim that Citus could replace Cassandra for high-velocity writes is a stretch. CAP triangle diagram is nice. 11 slides is good for timing.
GitHub Repository	**4**	**3.5**	Excellent repo organization: parts 01-04 all present with Docker configs, init scripts, custom ETL scripts, and docx submissions. All 5 team members have commits (Aisha: 3, An Chu: 4, Aurelia: 3, Sai Han Shan: 1, Mohammed: 1). Prof-tcsmith has access. Presentation PPTX included. **Issue:** 2 of 5 members have only 1 commit each — while technically meeting the minimum, this suggests minimal Git participation. -0.5 for weak commit distribution.

Part 4 Total: 39.5/40

GitHub Repository Assessment

Criterion	Status
Repo created and organized by parts	Yes -- excellent structure with full Docker configs
Prof-tcsmith invited as collaborator	Yes (origin remote = prof-tcsmith fork)
All team members have >= 1 commit	**Yes** -- all 5 members have commits
Part 4 slide deck in repo	Yes (LeftsideUndergrads_midterm_part04.pptx)
Submission docs present	Yes -- .docx for all 3 parts + custom ETL scripts
Modified ETL scripts included	Yes -- separate named scripts for parts 2 and 3
Additional materials	Full Docker infrastructure, init scripts, diagrams, READMEs

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Summary

Component	Points Available	Score
Part 1 -- Operational Databases & ETL	20	20
Part 2 -- Sharding	20	20
Part 3 -- Cassandra	20	20
Part 4 -- Repository	4	3.5
Part 4 -- Presentation delivery (in-class)	36	36
Total	**100**	**99.5/100**

Strengths

- Outstanding repository organization -- one of the best structured submissions
- Full Docker infrastructure included (not just submission docs) -- goes above and beyond
- All 5 team members contributed commits
- Custom ETL scripts are clean, well-documented, and correctly implemented
- The Part 2 ETL includes thoughtful dict-based deduplication for merged reference data
- Part 3 ETL correctly handles Cassandra date conversion with `.date()` call
- Comprehensive README with business case narrative

Areas for Improvement

- Some commit messages are generic ("Add files via upload") -- use descriptive messages
- Could benefit from more granular commits showing iterative work